

Haoyang Pei

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EDUCATION

New York University

Doctor of Philosophy in Electrical and Computer Engineering, Tandon School of Engineering

Sep 2022 – Present
New York, United States

New York University

Master of Science in Computer Engineering, Tandon School of Engineering

Sep 2020 – May 2022
New York, United States

North China Electric Power University

Bachelor of Engineering in Electrical Engineering and Automation

Sep 2013 – Jun 2017
China

RESEARCH INTERESTS

AI for healthcare and medical imaging, MRI reconstruction and denoising, multimodal learning, and generative models.

PROFESSIONAL EXPERIENCE

Microsoft

Applied Scientist Intern, Microsoft Research & Microsoft AI Health Team

May 2026 – Aug 2026
Redmond, WA, USA

- Developed a scalable foundation model for enhancing low-quality MRI images across diverse acquisition settings.
- Trained the model on a large-scale Kubernetes cluster equipped with NVIDIA B200 GPUs.
- Deployed pretrained models to a portable healthcare device via Microsoft Azure, enabling real-time inference.

NYU Langone Health & NYU Video Lab, New York University

Research Assistant, Center for Advanced Imaging Innovation and Research (CAI²R)

Feb 2023 – Present
New York, NY, USA

- Built an all-in-one self-supervised reconstruction framework for real-time 4D radial MRI across organs, resolutions, and dynamics.
- Developed a novel hybrid learning strategy for low-field MRI reconstruction and denoising without high-quality references.
- Built deep-learning frameworks for quantitative MRI, accelerating image acquisition and parameter fitting.
- **Project portfolio:** haoyangpei.github.io/projects.

BMEII, Icahn School of Medicine at Mount Sinai

Research Assistant, Biomedical Engineering and Imaging Institute (BMEII)

Jun 2022 – Feb 2023
New York, NY, USA

- Designed generative models for perfusion MRI mapping, enabling a 50% reduction in contrast agent dose.
- **Project portfolio:** haoyangpei.github.io/projects.

NYU Video Lab, New York University

Research Assistant, NSF Smart and Connected Communities (S&CC) Project

Jun 2021 – Aug 2022
New York, NY, USA

- Contributed to the development of a wearable navigation system for people with visual impairments.
- Evaluated how object distance, spatial resolution, and video compression rate affect object detection accuracy, identifying optimal settings for wireless transmission and edge computing.
- Developed transformer models for absolute depth estimation from monocular images, achieving a mean absolute error of 0.39 m for objects within 10 m.
- **Project portfolio:** haoyangpei.github.io/projects.

State Grid Corporation of China

Electrical Engineer, Shanxi Branch

Aug 2017 – Aug 2021
China

- Monitored and operated the citywide power grid serving 369,361 residents.
- Built a web platform to analyze operational data from 118 10-kV power lines and 14 substations.

TEACHING EXPERIENCE

Tandon School of Engineering, New York University

Teaching Assistant, ECE 6123 – Image and Video Processing (Spring 2022)

Feb 2022 – May 2022
New York, United States

SELECTED PUBLICATIONS

Book Chapter

1. Map the image: Machine learning for quantitative MR mapping. In *Advances in Magnetic Resonance Technology and Applications*, Vol. 13, pp. 275–302. Academic Press, 2025. [Book]
Li Feng, Haoyang Pei, Fang Liu

Journal Articles [Full List]

1. Hybrid Learning: A Combination of Self-Supervised and Supervised Learning for Joint MRI Reconstruction and Denoising in Low-Field MRI. *Physics in Medicine & Biology*. [Paper]
Haoyang Pei, Ding Xia, Xiang Xu, William Moore, Yao Wang, Hersh Chandarana, Li Feng
2. Deep Learning-Based Generation of DSC MRI Parameter Maps Using DCE MRI Data. *American Journal of Neuroradiology*. [Paper]
Haoyang Pei, Yixuan Lyu, Sebastian Lambrecht, Doris Lin, Li Feng, Fang Liu, Paul Nyquist, Peter van Zijl, Linda Knutsson, Xiang Xu
3. DeepEMC-T2 Mapping: Deep Learning-Enabled T2 Mapping Based on Echo Modulation Curve Modeling. *Magnetic Resonance in Medicine*. [Paper]
Haoyang Pei, Timothy M Shepherd, Yao Wang, Fang Liu, Daniel K Sodickson, Noam Ben-Eliezer, Li Feng
4. Rapid 3D T1 Mapping Using Deep Learning-Assisted Look-Locker Inversion Recovery MRI. *Magnetic Resonance in Medicine*. [Paper] **(August Editor Picks)**
Haoyang Pei, Ding Xia, Xiang Xu, Yang Yang, Yao Wang, Fang Liu, Li Feng

Conference [Full List]

1. A Unified All-in-One Self-Supervised Model for Accelerated 4D Radial MRI Across Organs, Resolutions, and Dynamics. *Proceedings of the 35th Annual Meeting and Exhibition of the ISMRM*. 2026. **(Oral, Summa Cum Laude, AMPC Selection Award)**
Haoyang Pei, Jingjia Chen, Yao Wang, Hersh Chandarana, Li Feng
2. Deep Learning-Optimized GRASP-Pro Reconstruction for Highly-Accelerated DCE-MRI. *Proceedings of the 34th Annual Meeting and Exhibition of the ISMRM*. 2025. **(Oral, Magna Cum Laude)**
Haoyang Pei, Li Feng
3. Hybrid Learning: A Novel Combination of Self-Supervised and Supervised Learning for MRI Reconstruction Without High-Quality Training Reference. *Proceedings of the 34th Annual Meeting and Exhibition of the ISMRM*. 2025. **(Oral)**
Haoyang Pei, Ding Xia, Xiang Xu, William Moore, Yao Wang, Hersh Chandarana, Li Feng
4. DeepGrasp4D: A General Framework for Highly-Accelerated Real-Time 4D Golden-Angle Radial MRI Using Deep Learning. *Proceedings of the 33rd Annual Meeting and Exhibition of the ISMRM*. 2024. **(Oral, Magna Cum Laude)**
Haoyang Pei, Hersh Chandarana, Daniel K Sodickson, Li Feng
5. Understanding the Impact of Image Quality and Distance of Objects on Object Detection Performance. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 11436–11442). IEEE.
Yu Hao, Haoyang Pei, Yixuan Lyu, Zhongzheng Yuan, John-Ross Rizzo, Yao Wang, Yi Fang
6. Rapid 3D T1 Mapping Using Deep Learning-Assisted Look-Locker Inversion Recovery MRI. *Proceedings of the 32nd Annual Meeting and Exhibition of the ISMRM*. 2023. **(Oral, Magna Cum Laude)**
Haoyang Pei, Ding Xia, Xiang Xu, Yang Yang, Yao Wang, Fang Liu, Li Feng
7. DSC-Derived Perfusion Map Generation from DCE MRI Using Deep Learning. *Proceedings of the 32nd Annual Meeting and Exhibition of the ISMRM*. 2023. **(Oral, Summa Cum Laude, AMPC Selection Award)**
Haoyang Pei, Yixuan Lyu, Sebastian Lambrecht, Doris Lin, Li Feng, Fang Liu, Paul Nyquist, Peter van Zijl, Linda Knutsson, Xiang Xu

AWARDS

- ISMRM: Summa Cum Laude (top 5%; 2023 and 2026); Magna Cum Laude (top 15%; 2023–2025)
- ISMRM Study Group Abstract Awards: 2nd Place, AI/ML, 2026; 1st Place, Low Field, 2025
- Dr. Li's Publication Award, NYU Tandon, 2025

SKILLS

Programming & Machine Learning: Python, MATLAB, PyTorch, NumPy, computer vision, transformer models, generative models, self-supervised learning, distributed training, Weights & Biases

Infrastructure & Deployment: Microsoft Azure, Kubernetes, Docker

Design & Presentation: PowerPoint (seven years of commercial presentation design experience), Adobe Photoshop, Adobe Lightroom